09-08 第2篇

阅读理解（说明文） How Does a Tiny Ant Find Its Way Home?

① If you watch ants on a summer afternoon, you will notice something strange and beautiful. A small black ant leaves its nest, zigzags along the ground, finds a drop of sugar on a paving stone, picks up a piece that is **twice its own weight**, and then — without a map, without Wi-Fi, without a single GPS satellite — walks **straight back** to its nest, often entering the tiny hole at the first try. How does such a small animal, with a brain no bigger than a grain of sand, manage to do this every day? Scientists have asked the same question for more than a hundred years, and what they have found is even stranger than you might think.

② The most famous "ant tool" is probably the **chemical trail**. When an ant finds food, it walks back to the nest, quietly leaving a thin line of a special chemical called a **pheromone**. Other ants pick up the smell with their **two long antennae** and follow it. The more ants walk along a good trail, the stronger the smell becomes, and the more new ants follow. This is how you sometimes see a perfect straight line of ants leading to a forgotten biscuit behind a sofa.

③ However, a pheromone trail alone cannot explain everything. Some ants live in **deserts**, where the wind blows pheromones away in minutes, yet these ants still travel hundreds of metres in a straight line to find food, and walk **directly back home** afterwards. One famous example is the **Saharan desert ant**, studied by scientists in the 20th century. These ants run over hot sand that is too dangerous to pause on; they must be fast, and they must get home without getting lost. So they use other clues.

④ One important clue is the **sun** — or more exactly, the **direction of sunlight in the sky**. The ant's eyes can see a kind of pattern in sunlight that humans cannot see without special glasses: **the polarisation of light**. This pattern is like a soft invisible arrow pointing to where the sun is. Even when part of the sky is covered by a cloud, the ant can still read the remaining pattern and find its "north" or "south" on the hot ground. Think of it as a natural **sun compass** built into the ant's eyes.

⑤ A second clue is something that sounds almost funny: **counting steps**. Scientists in Europe trained Saharan desert ants to walk along a straight path to food. Then, before the ants returned to the nest, the researchers did something very clever: some ants got **tiny stilts glued to their legs** (making the legs longer), and others had the tips of their legs **gently trimmed** (making them shorter). The ants with longer legs walked too **far**, and the ants with shorter legs stopped too **short**. This showed that the ants were

counting the number of steps they had taken on the way out, and walking the **same** number of steps back.

⑥ A third clue is the **landmark**. Many ants also remember key "landmarks" near their nest — a stone, a dry root, a pattern of grass. When they come close, these landmarks help them check and adjust their route, the way you use a red-roofed building to know that your school gate is near. Young ants often make **short practice trips** around the nest to learn these landmarks before they go on long food missions.

⑦ What if an ant still gets lost? Surprisingly, nature has a **backup plan**. Lost desert ants do not just wander forever. They start walking in a **spiral** of slowly growing circles around their last known point. If their nest is within this spiral, they find it. If it is not, the ant increases the search area step by step. This simple but patient method has been shown in many field studies and even copied by engineers for small search-and-rescue robots.

⑧ All these tools — the **pheromone trail**, the **sun compass**, the **step counter**, the **landmark memory**, and the **spiral search** — work together in a very small head. None of them is perfect on its own, but as a **team of simple rules**, they turn a tiny insect into a kind of walking navigation computer. In fact, some of the first ideas behind the maps inside our modern phones were partly inspired by studying how social insects find their way.

⑨ So next time you see a line of ants crossing the kitchen floor towards a fallen crumb, kneel down for a moment. That small black traveller is doing quiet, difficult work — reading sunlight, counting steps, trusting a chemical from a sister, and remembering a stone she saw only once. She is not simply "an ant". She is one of the oldest, quietest explorers on our planet, walking home with a piece of your biscuit.

注释： pheromone 信息素；antenna / antennae 触角；polarisation (光的)偏振；compass 罗盘；stilts 长高的支脚；landmark 地标

根据文章内容，选择最佳答案。

1. What is the main function of the pheromone trail, according to paragraph 2?

A. To fight other ants in a war.
B. To mark a path from a food source so other ants can follow the smell with their antennae.
C. To build a small nest underground.
D. To scare away the sun on hot days.

2. Why can desert ants, such as the Saharan desert ant, not rely mainly on pheromone trails?

A. Because deserts have too many other ants.
B. Because the wind blows pheromones away in minutes.
C. Because desert ants cannot smell at all.
D. Because there is no food in the desert.

3. What did scientists prove by gluing tiny stilts onto some ants' legs and trimming the legs of other ants?

A. That ants cannot walk without help.
B. That ants count the number of steps they take, and walk the same number back.
C. That ants are afraid of humans.
D. That all ants carry GPS devices.

4. According to paragraph 7, what do lost desert ants typically do to find their nest again?

A. Lie down and wait for rain.
B. Walk forever in a single straight line.
C. Walk in a slowly growing spiral of circles around their last known point.
D. Call other ants on a small phone.

5. What is the main purpose of this passage?

A. To scare readers away from all insects.
B. To teach readers how to build an ants' nest.
C. To show that ants use several simple rules that together form a clever way of finding their way home, and to invite us to respect these tiny navigators.
D. To explain why people should keep ants as pets.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "leaving a thin line of a special chemical called a pheromone. Other ants pick up the smell with their two long antennae and follow it"
【解析】信息素道路吸引其他蚂蚁沿路跟随。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "where the wind blows pheromones away in minutes"
【解析】大风使信息素迅速消散。A、C、D 均无依据。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段 "ants with longer legs walked too far, and the ants with shorter legs stopped too short ... the ants were counting the number of steps"
【解析】长腿蚂蚁走远了，短腿蚂蚁走近了，证明它们会"数步数"。A、C、D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑦段 "They start walking in a spiral of slowly growing circles"
【解析】迷路的沙漠蚂蚁采用扩大螺旋搜索。A、B、D 均错。故选 C。

5.
【答案】C
【题型】主旨大意题
【定位】第⑧—⑨段 "a team of simple rules" "one of the oldest, quietest explorers on our planet"
【解析】文章旨在介绍多种机制如何共同使蚂蚁成为小小导航者，并引发敬意。A、B、D 均不合文意。故选 C。